Predicting Fast-Growing Firms: Summary Report

Data Analysis 3 - Assignment 2
Submitted by: Balint Decsi (ID: 2506626), Zariza Chowdhury (ID: 2500086)
`https://github.com/zarizachow/Data-Analysis-3`

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Executive Summary

This report outlines the development and evaluation of predictive models designed to identify fast-growing firms using the Bisnode dataset. Our analysis defines "fast growth" as firms achieving the top 10% of sales growth year-over-year (2013 vs. 2012).

Key Decision: We recommend deploying the **Random Forest** model. It significantly outperforms traditional logistic regression benchmarks in identifying high-potential firms, reducing the expected business loss by approximately 20% under our specified cost structure.

Target Definition & Business Problem

We define a fast-growing firm as one in the top decile (top 10%) of log sales growth from 2012 to 2013. The distribution of firm growth is shown below.

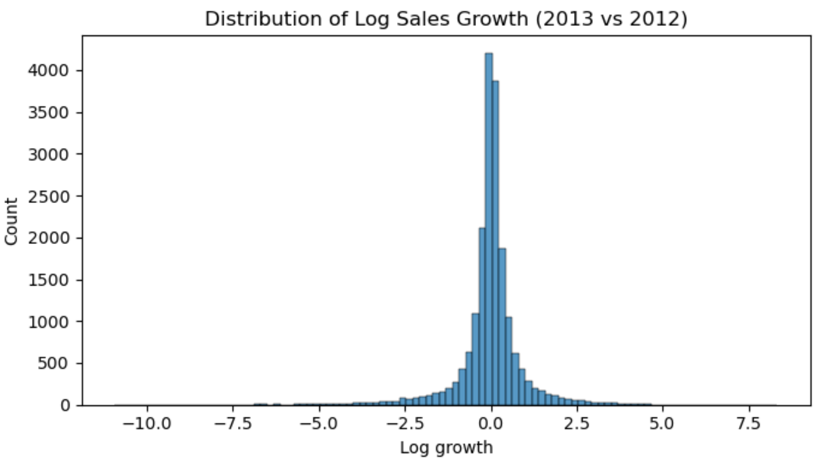

Figure 1: Distribution of Log Sales Growth (2012-2013)

- **Why this target?** Focusing on the top decile isolates the truly high-performers from steady-state firms. A one-year horizon (2012-2013) allows for immediate actionable insights compared to longer horizons which may introduce survivorship bias or noise.
- **Business Objective:** The goal is to invest in or target these firms early. Missing a fast grower (False Negative) is considered significantly more costly than investigating a false lead (False Positive).

- **Loss Function:** We assigned a cost ratio where the cost of a False Negative (FN) is **4 times** that of a False Positive (FP).

$$\text{Loss} = 1 \times \text{FP} + 4 \times \text{FN}$$

Model Performance Comparison

We evaluated three models: Logistic Regression (L2 regularization), LASSO (L1 regularization), and Random Forest. The non-linear nature of growth drivers, such as firm age, is a key reason for using flexible models like Random Forest.

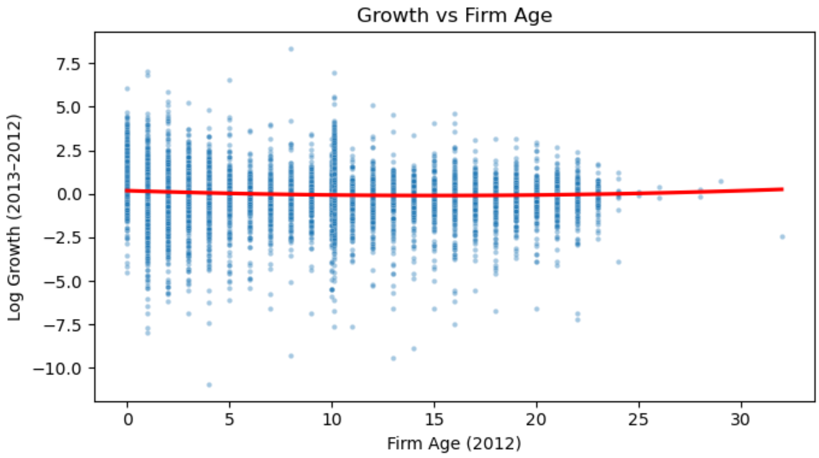

Figure 2: Growth vs. Firm Age: Younger firms show higher but more volatile growth.

Table 1: Model Performance (Probability Prediction)

Model	ROC-AUC	Avg Precision	Brier Score
Random Forest	**0.828**	**0.460**	**0.078**
Logit (LASSO)	0.753	0.269	0.121
Logit (L2)	0.753	0.269	0.121

The **Random Forest** model dominates the linear models. An AUC of 0.828 indicates strong discriminatory power. Notably, the Average Precision (0.460) is nearly double that of the Logit models, meaning that when the model predicts a firm is fast-growing, it is much more likely to be correct.

Decision Threshold & Expected Loss

To minimize business loss (prioritizing the capture of fast growers), we optimized the classification threshold for each model.

Table 2: Optimal Classification Performance (Lower is Better)

Model	Avg Expected Loss	Optimal Threshold
Random Forest	**0.255**	**0.150**
Logit (L2)	0.316	0.446

The Random Forest model yields an average expected loss of **0.255**, which is substantially lower than the Logit model's 0.316. The optimal probability threshold is approximately **15%**. This means if the model assigns a probability of fast growth greater than 15%, we should classify the firm as a target. This low threshold reflects our preference to avoid missing opportunities (high cost of False Negatives).

Industry Analysis

We validated the model across two key sectors: **Manufacturing** and **Services** (Repair, Accommodation, Food).

- **Manufacturing:** AUC = 0.817 | Expected Loss = 0.239
- **Services:** AUC = 0.838 | Expected Loss = 0.257

The model is robust across sectors, performing slightly better in the Services sector in terms of ranking (AUC) but incurring slightly higher loss due to sector-specific difficulty. It is safe to deploy the model in both domains.

Conclusion & Recommendation

We recommend implementing the **Random Forest** model for targeting fast-growing firms.

1. **High Accuracy:** It captures non-linear relationships (e.g., between age/size and growth) that linear models miss.
2. **Cost Effective:** It minimizes the weighted business loss effectively.
3. **Actionable:** Using a probability threshold of 15%, the sales/investment team can create a prioritized list of high-potential targets.